

KeepKey Firmware Test Report

Firmware 7.16.0 | 2026-09-17 20:30 | 12/12 PASSED

Scope: this report is a curated catalog of 12 tests. The CI run collected 25 (0 of them native firmware unit tests); 0 SKIPPED and did not execute, usually because the emulator predates the firmware the test targets -- a skip is not evidence the feature works. Absence from this report is NOT evidence that a feature is untested -- check the JUnit artifacts.

Sections

ER ERC-7730 v2 Certified Clear-Signing Conformance [NEW] -- 12/12 passed

ER. ERC-7730 v2 Certified Clear-Signing Conformance [NEW]

ERC-7730 definitions are compiled into a bounded canonical program, authenticated by KeepKey's certified delegation hierarchy, and interpreted against the exact calldata or EIP-712 document being signed. The host supplies types, labels and formatting policy; it never supplies authoritative decoded transaction values. A certified-definition failure refuses signing and cannot silently downgrade to a less-specific certified display.

User Flow

AUTHENTICATION AND REPLAY BOUNDARY:

- K773 envelope: version + purpose + canonical C773 program + sorted Merkle proof + 139-byte KeepKey delegation certificate + compact secp256k1 signature.
- Purpose digest: SHA256("KEEPKEY:ERC7730:CATALOG\0" || catalog_root).
- Definition id commits to the complete envelope; offset-zero replay is re-hashed before staged interpreter output becomes authoritative.
- Catalog lookup binds kind, chain, target and selector/typeHash. Unknown, ambiguous, reordered, oversized and recursively repeated definitions fail closed.

CANONICAL PROGRAM AND DEVICE INTERPRETER:

- 179-byte header; sorted UTF-8 strings; canonical recursive ABI/EIP-712 node tree; typed structured/container/literal paths; typed literals and conditions; formatter, display, deployment/domain binding and resource sections.
- Recursive tuples and arrays through depth 12, 64 ABI nodes, 64 aggregate array elements, negative indices, half-open slices, nested array frames and separators.
- Display bytecode: intent, atomic interpolation fallback, fields, nested groups, arrays, embedded calls and explicit end; all forward/back links and depth are recomputed.
- Conditions: always, never, optional, empty/not-empty, in/not-in and must-match.
- Formatters: raw, native/token amounts, NFT, date, duration, unit, enum, chain, address name, ticker, ERC-7930 interoperable address, embedded calldata and encrypted fallback.
- Signed token/network records establish ticker, decimals and native currency. Live results may annotate but cannot replace a device-decoded value.

REAL-WORLD AND EXHAUSTIVE EVIDENCE:

- Exact KeepKey SDK THORChain router deposit calldata, including vault, asset, amount and memo, is selector-bound and firmware-validated.
- Official Uniswap Router02 signed transaction and Permit2 typed-data fixtures bind the expected selector, target, chain, typeHash and domain constraints.
- Every one of 1,450 calldata formats exposed after bounded include expansion in the official ERC-7730 registry compiles and reaches the firmware catalog verifier.
- Deep ParaSwap Augustus multiSwap/megaSwap definitions exercise depths 9 and 11 while the compacted signing workflow remains exactly 4,096 bytes of fixed SRAM.
- The SDK/Vault consumer path accepts the same signed catalog on ethSignTransaction and from Relay/ShapeShift quote payloads, preserves it through swap construction, preloads the primary envelope, and answers device-driven definition requests in 1,024-byte chunks. Missing, malformed, mismatched, oversized and over-depth requests abort.
- Consumer evidence: hdwallet PR #3 passed build/lint and 8 focused catalog/policy tests; keepkey-vault PR #444 passed 59 focused REST schema and real Relay swap parser tests, Linux/macOS release builds and the packaged Intel macOS smoke test before merge.

LIMIT OF THIS SECTION: catalog-verifier success ends at the expected UNTRUSTED result when tests use a synthetic certificate. A trusted-success emulator capture additionally requires a certificate issued by the KeepKey root for the test delegate.

Tests: 12/12 -- ALL PASSED

✓ER1 test_official_registry_all_calldata_formats_reach_firmware

All 1,450 official calldata formats reach firmware (test_erc7730_compiler.py)
Recursively merges bounded includes, compiles every applicable registry format and feeds each complete canonical program through the firmware catalog verifier.

✓ER2 test_compiles_and_checks_exact_keepkey_sdk_thorchain_swap

Exact KeepKey SDK THORChain swap fixture (test_erc7730_compiler.py)
Checks selector, vault, asset, amount, dynamic memo and transaction value before firmware validation.

✓ER3 test_compiles_official_uniswap_tuple_fixture_through_firmware

Official Uniswap signed transaction (test_erc7730_compiler.py)

Binds the real Router02 target, chain, selector and published transaction hash.

✓ER4 test_compiles_official_uniswap_eip712_fixture_through_firmware

Official Permit2 EIP-712 fixture (test_erc7730_compiler.py)

Proves canonical encodeType/typeHash, domain constraints and signed token/network metadata reach the firmware parser.

✓ER5 test_loads_bounded_includes_and_compiles_array_backed_group

References, includes and grouped arrays (test_erc7730_compiler.py)

Exercises deterministic include merge, shared field references and a group executed inside an authenticated array frame.

✓ER6 test_compiles_recursive_array_iteration_frames

Recursive ABI arrays and separators (test_erc7730_compiler.py)

Builds nested array begin/end links and validates their declared resource depth.

✓ER7 test_compiles_interpolated_intent_and_metadata_enum

Atomic interpolation and enum maps (test_erc7730_compiler.py)

Every interpolated operand is also an unconditional field; failure selects the fallback intent atomically.

✓ER8 test_compiles_typed_if_not_in_and_must_match_conditions

Typed visibility and must-match conditions (test_erc7730_compiler.py)

Typed literal sets control visibility; must-match is a signing assertion.

✓ER9 test_signed_catalog_envelope_commits_program_proof_and_certificate

Certified Merkle catalog envelope (test_erc7730_catalog.py)

Verifies deterministic root construction, purpose digest, signature and exact wire layout.

✓ER10 test_catalog_refuses_unknown_ambiguous_and_malformed_requests

Catalog lookup fails closed (test_erc7730_catalog.py)

Unknown selectors, oversized chunks, excessive recursion and ambiguous tuples are rejected.

✓ER11 test_definition_chunk_round_trip_is_byte_exact

Canonical protocol transport is byte exact (test_erc7730_protocol_bindings.py)

Protobuf bindings preserve definition id, offsets, total length and envelope bytes.

✓ER12 test_token_amount_without_token_uses_firmware_raw_fallback

Unknown tokens never become trusted labels (test_erc7730_compiler.py)

A missing token reference displays the device-decoded raw integer and forbids token-specific auxiliary metadata.